

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CIVICS GROUP

INDEX NUMBER

Mathematics

9758/01

Paper 1

24 August 2018

3 hours

Additional materials: Answer Paper
 Cover Page
 List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number on the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.



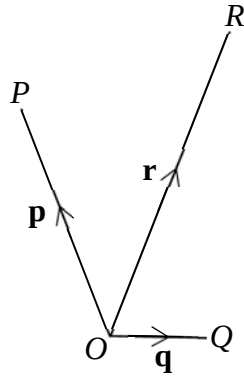
- 1 The variables x and y are related by

$$\frac{dy}{dx} = \sqrt{y^2 + e^{2x}}.$$

It is given that the curve of y passes through the point $(0,1)$. Find the Maclaurin series for y in ascending powers of x , up to and including the term in x^2 . Give the coefficients in exact form. [4]

- 2 Find $\int_0^\pi x^2 \sin(nx) dx$, where n is an even integer. Leave your answer in the form $\frac{k\pi^2}{n}$, where the value of k is to be determined. [4]

3



The origin O and the points P , Q and R lie in the same plane, where $\vec{OP} = \mathbf{p}$, $\vec{OQ} = \mathbf{q}$ and $\vec{OR} = \mathbf{r}$ (see diagram).

- (i) Explain why $\mathbf{r}$ can be expressed as $\mathbf{r} = \alpha\mathbf{p} + \beta\mathbf{q}$, for constants α and β . [1]
- (ii) The point X is on PR such that $PX : XR = 2 : 1$. It is given that the area of triangle OPX is equal to the area of triangle OQR , find the ratio $\alpha : \beta$ in the case where α and β are positive. [4]
- 4 (a) Solve the inequality $x^2(x-5) \geq (x-5)(2kx-k^2)$, given that k is a constant and $k < 5$. [3]
- (b) It is given that $f(x) = a + bx - x^2$ and $g(x) = |x - c|$ where a , b and c are constants and $2 < c < 4$. Given further that $f(2) = g(2)$, $f(5) = g(5)$ and $f(4) - g(4) = \frac{8}{3}$, find the values of a , b and c . [3]
- Hence, find the exact value of the area bounded by the graphs of $y = f(x)$ and $y = g(x)$ for $2 \leq x \leq 3$. [3]

- 5 (a) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = \frac{1}{n!} \text{ and } u_{n+1} = u_n - \frac{1}{(n-1)! + n!} \text{ for } n \geq 1.$$

(i) Find a simplified expression for $\sum_{r=1}^N \frac{1}{(r-1)! + r!}$. [2]

(ii) Hence show that $\sum_{r=4}^N \frac{1}{(r-1)! + r!} < \frac{1}{24}$. [2]

- (b) D'Alembert's ratio test states that a series of the form $\sum_{r=0}^{\infty} a_r$ converges when

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1, \text{ and diverges when } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1. \text{ When } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1, \text{ the test}$$

is inconclusive. Using the test, explain why the series $\sum_{r=0}^{\infty} \frac{(-1)^r \pi^{2r}}{(2r)!}$ converges and state the sum to infinity of this series. [4]

- 6 It is given that $f(x) = \frac{x}{\sqrt{2+x^2}}$.

(i) Find $\int_0^k f(x) dx$ in terms of k , where k is a positive constant. [2]

- (ii) It is now given that $g(x)$ is the first three terms, in ascending powers of x , of the series expansion of $f(x)$. Find $g(x)$ and the set of values of x for the expansion to be valid. [5]

(iii) Given that $\left| \int_0^k f(x) dx - \int_0^k g(x) dx \right| < 0.005$, where $k > 0$, find the range of values of k correct to 4 significant figures. [2]

- 7 Two swimmers are training for a long distance swimming competition. They are to swim a distance of 4 km by swimming 80 laps at a swimming pool, where 1 lap covers 50 m from one end of the pool to the other end. Both swimmers aim to complete the distance in between $2\frac{1}{3}$ hours and $2\frac{5}{6}$ hours inclusive.

- (i) Swimmer A swims the first lap in T seconds and each subsequent lap takes 1.5 seconds longer than the previous lap. Find the set of values of T which will enable A to complete the distance within the required time interval. [3]
- (ii) Swimmer B swims the first lap in t seconds and the time for each subsequent lap is 1.5% more than the time for the previous lap. Find the set of values of t which will enable B to complete the distance within the required time interval. Leave your answers correct to 2 decimal places. [3]
- (iii) Assuming each swimmer completes a distance of 1.8 km in exactly 50 minutes, determine which swimmer is faster in their 80th lap. Justify your answer. [4]

- 8 (a) Without using a calculator, find the complex numbers z and w which satisfy the simultaneous equations

$$(3+i)z + 3w = -5i \quad \text{and} \quad (i-2)z - 6iw = 1-3i. \quad [4]$$

- (b) (i) Given that $-\frac{1}{2}(1+i)$ is a root of the equation

$$k\omega^4 - 2\omega^3 + 5\omega^2 + 6\omega + 4 = 0,$$

find the value of the real number k and the other roots in exact form. [5]

- (ii) The roots of the equation in part (i) are denoted by ω_1 , ω_2 , ω_3 and ω_4 , where $\arg \omega_1 < \arg \omega_2 < \arg \omega_3 < \arg \omega_4$.

Find $\frac{\omega_3}{\omega_4}$ in polar form $r(\cos \theta + i \sin \theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$.

Give r and θ in exact form. [3]

- 9** The curve C has equation $y = f(x)$, where

$$f(x) = \frac{2x^2 + kx + 8}{x - 4} \text{ and } k \text{ is a constant.}$$

- (i) Find the equation of the tangent to C at the point where $x = 1$. [4]
 (ii) Find the range of values of k for which C has more than one stationary point. [2]

Let $k = -7$ for the remaining parts of the question.

- (iii) Sketch C , stating the coordinates of the turning point(s) and the points of intersection with the axes and the equations of any asymptotes. [3]
 (iv) On separate diagrams, sketch the graphs of
 (a) $y = f(|x|)$, [2]
 (b) $y = f'(x)$, [2]
 stating the equations of any asymptotes and the coordinates of any points of intersection with the axes where appropriate.

- 10** A curve C_k has parametric equations

$$x = 1 + k \cos \theta, \quad y = -2 + \frac{1}{2}k \sin \theta,$$

where k is a positive constant.

- (i) Find the cartesian equation of C_k and show that its gradient function is $\frac{1-x}{4(y+2)}$. [4]
 (ii) On the same diagram, sketch the graphs of C_1 and C_4 . Label the two graphs clearly. [3]

On a map, the curves C_1 , C_2 , C_3 and C_4 represent the contours of a mountain. A stream flows down the mountain. Its path on the map is always at right angles to the contour it is crossing.

- (iii) Explain why the path of the stream is modelled by the differential equation

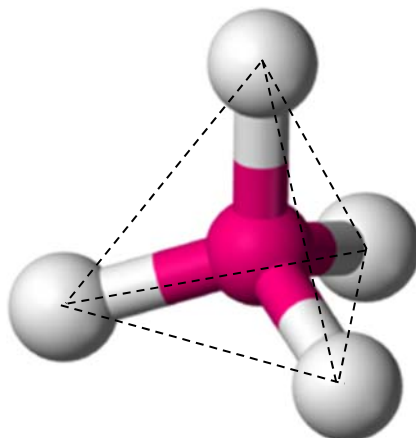
$$\left(\frac{1}{y+2} \right) \frac{dy}{dx} = \frac{4}{x-1}.$$

By considering $\int \frac{1}{y+2} dy = \int \frac{4}{x-1} dx$, show that the path of the stream on the map is represented by the general solution $y = A(x-1)^4 - 2$, where A is an arbitrary constant. [5]

- (iv) The path of the stream on the map passes through the point $(-1, -1)$. find the equation of the path. [1]

- 11** In Chemistry, the molecular structure of chemical compounds is often of interest to chemists as this will aid them in predicting the chemical properties of the compound.

In studying the molecular structure of *silicon tetrachloride*, it is found that this compound takes the form of a regular tetrahedron, that is, it consists of a *silicon* atom at the centre with four *chlorine* atoms symmetrically positioned at the corners of the regular tetrahedron (see diagram).



Suppose the centers of the *chlorine* atoms are at the points A , B , C and D with coordinates $(5, -2, 5)$, $(5, 4, -1)$, $(-1, -2, -1)$ and $(7, -4, -3)$ respectively, where $ABCD$ forms a regular tetrahedron.

- (i) Verify that triangle ABC is an equilateral triangle. [2]
- (ii) Find a vector that is perpendicular to the plane containing triangle ABC . [1]
- (iii) π_1 is a plane that is perpendicular to $\overrightarrow{AB}$ and passes through the mid-point of the line segment AB . Find the cartesian equation of π_1 . [2]
- (iv) π_2 is a plane that is perpendicular to $\overrightarrow{BC}$ and passes through the mid-point of the line segment BC . Given that π_1 and π_2 meet in the line l , find a vector equation for l . [3]
- (v) The position of the *silicon* atom is at the point G , where G is equidistant from A , B , C and D . Find the coordinates of G . [3]
- (vi) The angle AGD is also known as the bonding angle of the compound. Find the bonding angle. Show your workings clearly. [2]